Outlook

ISPRS 2026 - Abstract Acceptance Notification

From XXV ISPRS Congress <isprs2026@conftool.com>
Date Sun 1/4/2026 9:38 PM
To Dr. Deni Suwardhi, S.T, M.T. <deni.suwardhi@itb.ac.id>
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Dear Deni Suwardhi,

Please find the review results of your abstract submission(s) below, including abstract submitted to ISPRS Archives, abstract-only contributions, and applied abstracts.

What's the next step?

If your abstract(s) are accepted for ISPRS Archives, please prepare the full paper by addressing all the comments from the reviewers and following the ISPRS Guidelines for authors preparing a full paper (https://www.isprs.org/documents/orangebook/app5.aspx). The deadline for submitting your camera-ready full paper is March 30, 2026. You will be able to upload your full paper after January 31, 2026.

For accepted abstract-only and applied-abstract submissions, a full paper is not required. However, in order for your submission to be included in the final program, authors of an accepted abstract are required to submit a properly formatted final revised version of the extended abstract by addressing reviewers' comments and following to the ISPRS Guidelines by March 30, 2026.

Please be reminded that at least one author of each accepted submission must register by the deadline of April 27, 2026 to be included in the program and/or proceedings; furthermore, a maximum of two (2) accepted submissions will be permitted per registered (full fee or reduced fee) co-author.

If you have not completed your congress registration, we encourage you to register as soon as possible. The early registration deadline ins March 2, 2026.

Contribution Details

ID: 1299
Title: Bridging Semantic Mesh, CityGML, and Gaussian Splatting for Urban Modelling and Visualization

Review Result of the Program Committee: This contribution has been accepted and will be published in the 2026 proceedings.
Your paper will be published in the ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences (https://www.isprs.org/publications/archives.aspx), subject to the final camera-ready version.

One of the co-authors has to be registered to the 2026 ISPRS Congress for final publication of your paper.

Overview of Reviews

Review 1
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Contribution of the Submission

The core contribution of this abstract is the proposal of a multi-model interoperability framework integrating Semantic Mesh, CityGML, and 3D Gaussian Splatting (3DGS), bridging the technical gap in urban modelling between semantic structure, geometric detail, and high-fidelity real-time visualisation. The solution overcomes the limitations of single-model approaches by achieving complementary strengths in semantics, geometry, and visualisation. It establishes a standardised, reusable technical chain encompassing UAV image processing, deep learning-based semantic extraction, multi-scale modelling, and 3DGS reconstruction. Through an ETL workflow and unified digital twin framework, it supports collaborative multi-model querying, evaluation, and cross-task adaptation. Preliminary testing has validated the models' complementary nature, delivering a solution for urban digital twins that balances standardisation, granularity, and immersive quality. This approach caters to diverse requirements including urban planning and environmental simulation.

Evaluation of the Contribution

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*Innovation                (15%): 6
Scientific formulation     (10%): 8
Experiments and validation (15%): 6
Relevance                  (10%): 8
Quality of Presentation    (10%): 6
Overall Recommendation     (40%): 8
Total points (out of 100)       : 72
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Comments for the authors

This paper focuses on the shortcomings of three core representation methods in urban modelling, proposing an integrated multi-model interoperability framework combining Semantic Mesh, CityGML, and 3D Gaussian Splatting (3DGS). This effectively addresses the challenge that single models struggle to simultaneously meet the diverse demands of semantic analysis, geometric simulation, and immersive visualisation. Preliminary testing has validated the functional complementarity and scenario adaptability of these three representation types. This framework demonstrates outstanding technological integration, linking a complete technical chain through ETL workflows to support unified spatial environment management. It exhibits strong engineering practicality and preliminary feasibility, with the overall results being commendable.
To deepen research on this foundation, current testing scenarios remain relatively concentrated within campus environments. Further validation is required regarding its adaptability to typical urban settings such as high-density urban areas. Comprehensive experiments are ongoing, with scope to supplement quantitative data on geometric consistency and enhance comparative analysis against mainstream multi-model solutions from recent years. Practical implementation details concerning processing efficiency and computational requirements for city-scale modelling also warrant further refinement.
Building upon the solid foundation established in this paper, it is recommended to expand testing scenarios to enrich validation dimensions, supplement relevant quantitative assessments and comparative analyses to more fully highlight advantages, appropriately refine implementation details for city-level applications, and simultaneously optimise data exchange mechanisms and semantic integration depth between models. These enhancements will better enable the solution to deliver practical value.

Review 2
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Contribution of the Submission
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This study develops a unified interoperability framework that integrates CityGML, Semantic Mesh
models, and 3D Gaussian Splatting, for analytical queries, simulation workflows, and photorealistic
visualization within a consistent multi-scale urban modelling pipeline by combining rich semantics,
detailed geometry, and high-fidelity rendering.


Evaluation of the Contribution
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*Innovation                (15%): 8
Scientific formulation     (10%): 8
Experiments and validation (15%): 8
Relevance                  (10%): 8
Quality of Presentation    (10%): 8
Overall Recommendation     (40%): 9
Total points (out of 100)       : 84


Comments for the authors
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The manuscript is good. Please add a brief model validation in both methods and results parts.
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Your organizers of XXV ISPRS Congress.

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XXV ISPRS Congress
Toronto, Canada | 4 -11 July 2026

Submission / registration website: https://conftool.com/isprs2026/
Congress website: https://www.isprs2026toronto.com/

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